**5** **(a)** **(i)** Waves from the two slits overlap and superpose (debrief point) at points on the screen B1

When path difference from slits to point is multiples of wavelength / phase difference between the two waves is multiples of $2\pi$, constructive interference gives bright fringe B1

**(ii)** separation $= \dfrac{(590\times10^{-9})(2.3)}{1.2\times10^{-3}}$ M1

$= 1.1$ mm or 1.13 mm A1

**(b)** $\sin\sin\theta = \frac{\lambda}{b} = \frac{590\times10^{-9}}{0.31\times10^{-3}} \approx \frac{x}{D}$ must see sinθ M1

$(\tan\tan\theta = \frac{x}{D} = \frac{x}{2.3}$ led to $x = \frac{(590\times10^{-9})(2.3)}{0.31\times10^{-3}}$

width = 2*x* M1

= 8.8 mm A0

**(c)** Diffraction minimum is (8.8 / 2 =) 4.4 mm from P and the fourth order bright fringe is (1.1 × 4 =) 4.4 mm from P B1

Position of 4th order interference maximum coincides/overlap with (first order) diffraction minimum. B1

**6 (a)**